THERMO SENSE: Power Budgeting on a Chip Using Thermal Modelling and Linear Programming

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***Abstract*—Thermal management is an important challenge in modern integrated circuits because localized power dissipation can produce hot spots and reduce reliability. This project presents THERMO SENSE, a computational framework for power budgeting on a chip using steady-state heat conduction modelling and linear programming. A two-dimensional chip domain of 10 mm $\times$ 10 mm is discretized into a 40×40 grid. Four internal heat-source regions are considered, with a total power budget of 5 W and fixed-temperature boundaries at 300 K. The heat equation is discretized using a finite-difference method, and the resulting sparse linear system is solved using SciPy. A thermal response matrix is then used to formulate a linear program that minimizes the maximum chip temperature while satisfying the total power constraint. The reported simulation reduces the peak temperature from 352.91 K under uniform power allocation to 342.01 K under optimized allocation. The results demonstrate how numerical heat transfer modelling and optimization can be combined to improve power distribution across a simplified chip domain.**

***Index Terms*—Thermal modelling, power budgeting, heat conduction, finite difference method, sparse linear systems, linear programming, chip thermal management.**

I. Introduction

Power dissipation in integrated circuits generates heat, and non-uniform power distribution can create localized temperature peaks. Excessive temperature may affect circuit performance, reliability, and operating lifetime. Consequently, thermal-aware power allocation is an important part of electronic system design.

A uniform distribution of power is simple to implement, but it does not necessarily minimize the maximum temperature. Different source locations can have different thermal influences because of their positions relative to the chip boundaries and other heat sources.

This project investigates thermal-aware power budgeting using a two-dimensional steady-state heat conduction model. The chip is represented by a finite grid, and four internal regions are treated as controllable heat sources. A numerical thermal solver predicts the temperature distribution, while a linear programming formulation selects a power allocation that minimizes the peak temperature under a fixed total power budget.

The main objectives are to construct a numerical thermal model, compare uniform and optimized power allocations, and demonstrate the use of constrained optimization for thermal management.

II. Mathematical Modelling

A. Steady-State Heat Equation

Fourier's law relates heat flux to the temperature gradient:

$$\mathbf{q} = -k\nabla T \tag{1}$$

where $\mathbf{q}$ is the heat flux, k is the thermal conductivity, and T is temperature.

For steady-state heat conduction with internal heat generation, the governing equation is

$$\nabla \cdot (k\nabla T) + Q = 0 \tag{2}$$

Assuming constant thermal conductivity, this becomes

$$\nabla^2 T = -\frac{Q}{k} \tag{3}$$

Here, Q represents volumetric heat generation in $\mathrm{W/m^3}$. The model assumes that temperature does not vary with time.

B. Chip Geometry and Boundary Conditions

The chip domain is a square measuring $10\,\mathrm{mm} \times 10\,\mathrm{mm}$. It is discretized into a 40×40 computational grid. The thermal conductivity is assumed to be $150\,\mathrm{W/(m\,K)}$, and the boundary temperature is fixed at $300\,\mathrm{K}$.

Four internal source regions are modelled, each occupying a 4×4 group of grid cells. The total available power is constrained to 5 W. A thickness of 1 mm may be used to convert between total source power and volumetric heat generation, provided that this assumption matches the implementation in the simulation notebook.

The boundary condition is expressed as

$$T(x, y) = 300\,\mathrm{K} \qquad \text{on the chip boundary.} \tag{4}$$

C. Finite-Difference Discretization

For a uniform grid with spacing h, the two-dimensional Laplacian is approximated using the five-point stencil:

$$\nabla^2 T_{i,j} \approx \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1} - 4T_{i,j}}{h^2}. \tag{5}$$

Substitution into the governing heat equation produces a system of linear equations for the unknown grid temperatures. The system is written as

$$\mathbf{AT} = \mathbf{b}, \tag{6}$$

where $\mathbf{A}$ represents the discretized differential operator, $\mathbf{T}$ contains the unknown temperatures, and $\mathbf{b}$ contains the source and boundary contributions.

The matrix is sparse because each interior grid point couples only to its immediate neighbours. The system is solved using the sparse linear solver provided by SciPy.

III. Thermal Response and Power Allocation

A. Thermal Response Matrix

The temperature response to the source powers can be represented in matrix form as

$$\mathbf{T} = T_{\mathrm{amb}}\mathbf{1} + \mathbf{Hp}, \tag{7}$$

where $T_{\mathrm{amb}} = 300\,\mathrm{K}$, $\mathbf{p}$ is the vector of source powers, and $\mathbf{H}$ is the thermal response matrix.

Each column of $\mathbf{H}$ describes the temperature response produced by a unit of power in one source region. The response matrix allows the optimizer to evaluate alternative power allocations without repeatedly solving the complete heat equation for every candidate.

B. Uniform Power Allocation

With four source regions and a total power budget of 5 W, the uniform allocation is

$$p_i = \frac{5}{4} = 1.25\,\mathrm{W}, \qquad i = 1, 2, 3, 4. \tag{8}$$

This allocation provides a baseline for evaluating the benefit of thermal optimization.

C. Peak-Temperature Minimization

The optimization objective is to minimize the maximum temperature across the grid. An auxiliary variable t represents the upper bound on every grid temperature.

The linear programming problem is

$$\min_{\mathbf{p},t} \quad t \tag{9}$$

$$\text{subject to} \quad T_i(\mathbf{p}) \le t, \quad \forall i, \tag{10}$$

$$\sum_{j=1}^{4} p_j = 5\,\mathrm{W}, \tag{11}$$

$$p_j \ge 0, \quad j = 1, 2, 3, 4. \tag{12}$$

Using the thermal response model, each temperature constraint is linear in the power variables. Therefore, the problem can be solved using a standard linear programming solver, such as `scipy.optimize.linprog`.

IV. Numerical Implementation

The project is implemented in Python using NumPy, SciPy, and Matplotlib within Google Colab.

The computational procedure consists of the following steps:

1) Define the chip geometry, grid resolution, material conductivity, and boundary temperature.
2) Identify the four internal heat-source regions.
3) Construct the finite-difference matrix for the steady-state heat equation.
4) Solve the sparse system to obtain the temperature response to the source powers.
5) Construct the thermal response matrix.
6) Solve the linear program for the optimized power allocation.
7) Compare the peak temperature and power allocation with the uniform-power baseline.

The primary software components are NumPy for numerical arrays, SciPy for sparse linear algebra and optimization, and Matplotlib for temperature plots and comparisons.

V. Results and Discussion

A. Power Allocation Results

The reported uniform and optimized allocations are summarized in Table I.

TABLE I
Comparison of Power Allocations

Source region	Uniform (W)	Optimized (W)
1	1.250	0.801
2	1.250	1.364
3	1.250	1.417
4	1.250	1.417
Total	5.000	4.999

The optimized solution reduces power in the first source region and increases the allocation to the other regions. This indicates that equal power allocation does not necessarily produce the lowest peak temperature. The optimizer redistributes the available power to reduce the maximum temperature predicted by the thermal model.

The displayed optimized values sum to 4.999 W because the values are rounded. The full-precision solver output should be used when checking the exact power constraint.

B. Peak Temperature Comparison

TABLE II
Peak Temperature Before and After Optimization

Condition	Peak temperature (K)
Uniform allocation	352.91
Optimized allocation	342.01
Reduction	10.90

The peak temperature decreases from 352.91 K to 342.01 K, corresponding to a reduction of approximately 10.90 K using the displayed values. This result shows that thermal-aware allocation can improve the predicted thermal performance without exceeding the total power budget.

The outcome depends on the assumed conductivity, source locations, fixed-temperature boundaries, and grid discretization. It should therefore be interpreted as a result for the modelled chip rather than as a universal reduction for all integrated circuits.

C. Temperature Distribution

Figure 1 compares the steady-state temperature distributions under uniform and optimized power allocation.

The uniform allocation produces a peak temperature of 352.91 K, whereas the optimized allocation reduces the peak to 342.01 K, a reduction of approximately 10.90 K. This comparison demonstrates the benefit of redistributing power to reduce the maximum predicted chip temperature.

VI. Project Pipeline and Team Contributions

The project was organized into six stages, connecting theoretical modelling with numerical implementation and optimization.

The division of responsibilities follows the six-stage project plan. The stages collectively cover physical modelling, mathematical analysis, numerical computation, optimization, and result verification.

VII. Limitations and Future Work

The current model uses a constant thermal conductivity and fixed-temperature boundary conditions. These assumptions simplify the computation but do not capture all the thermal behaviour of a practical chip package.

The source regions are represented using idealized rectangular grid areas. The finite grid resolution can also introduce discretization error, and the steady-state formulation does not describe transient temperature changes during workload switching.

Future work could include grid-convergence analysis, temperature-dependent material properties, more realistic package and cooling boundary conditions, and transient thermal simulation. The optimization could also incorporate individual source limits or additional practical constraints.

VIII. Conclusion

This project presents a computational method for thermal-aware power budgeting on a simplified chip domain. A two-dimensional steady-state heat equation is discretized using the finite-difference method and solved as a sparse linear system. The resulting thermal response matrix is used to formulate a linear program that minimizes the maximum grid temperature under a fixed total power budget.

For the reported simulation, optimized power allocation reduces the peak temperature from 352.91 K to 342.01 K, a decrease of approximately 10.90 K. The results illustrate the potential value of combining numerical heat-transfer analysis with constrained optimization. Further validation with realistic material, package, and cooling models is needed before applying the approach to a physical chip.

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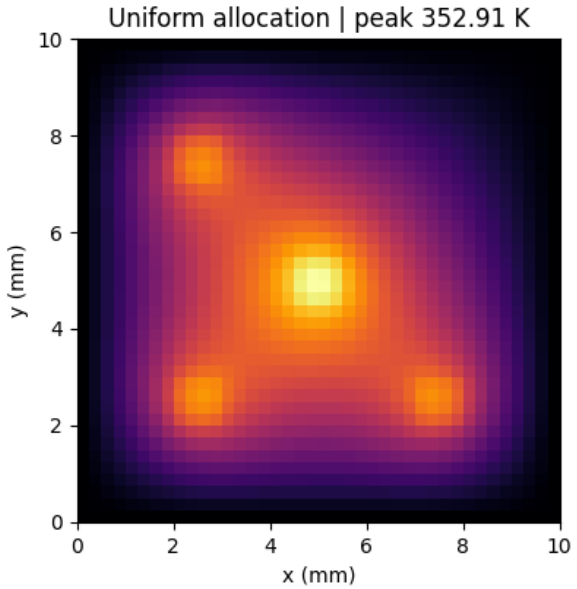
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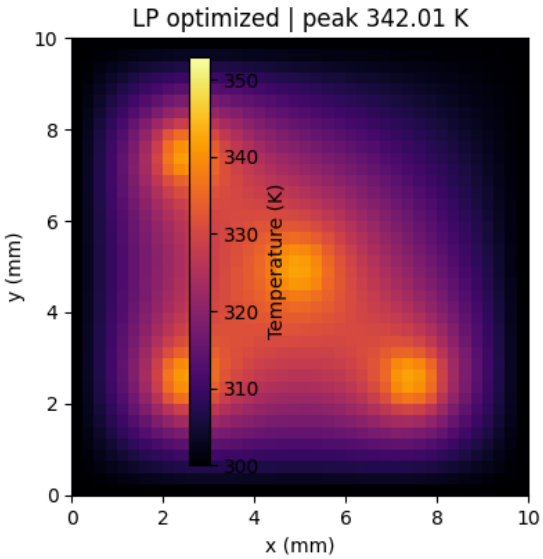
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(a) Uniform power allocation



(b) LP-optimized power allocation

Fig. 1. Steady-state temperature distributions for uniform and LP-optimized power allocation. The peak temperature decreases from 352.91 K to 342.01 K. Both plots use the same temperature colour scale.

TABLE III
SIX-STAGE PROJECT PIPELINE AND TEAM RESPONSIBILITIES

Stage	Topic	Main activity	Assigned member
1	Physical model	Define the chip domain, material properties, and steady-state heat equation.	Muhammed Yaseen
2	Mathematical formulation	Apply boundary conditions and derive the finite-difference system.	Muhammed Yaseen
3	Numerical solution	Build and solve the sparse thermal model.	Sharika V
4	Thermal response	Construct the source-to-temperature response representation.	Parvathy MS
5	Optimization	Formulate and solve the peak-temperature minimization problem.	Devika S Kumar
6	Verification and reporting	Compare allocations, evaluate results, and prepare the final report.	Bhagath CR